

Practice Questions for Basic Variable

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1 Questions

1. In the linear program presented on Slide 1, what are the basic variables when x_1 and x_2 are set to 0?
 - a. x_1, x_2, w_1
 - b. w_1, w_2, w_3
 - c. x_1, w_2, w_3
 - d. w_1, x_2, w_3
 - e. x_1, x_2, x_3
2. In the context of the Simplex Method, what does it mean for a variable to be "basic"?
 - a. A variable that is always set to zero in the simplex tableau.
 - b. A variable that is used to convert inequality constraints into equality constraints.
 - c. A variable whose value is determined by the values of other variables in the system of equations.
 - d. A variable that has a positive coefficient in the objective function.
 - e. A variable that is always present in the final optimal solution.
3. Why is the concept of basic and non-basic variables crucial in the simplex method?
 - a. It simplifies the calculation of the objective function.
 - b. It allows for a systematic exploration of the feasible region's vertices.
 - c. It ensures the non-negativity of all variables.
 - d. It guarantees the convergence of the algorithm.
 - e. It helps in identifying the constraints that are binding.
4. How does the identification of basic and non-basic variables relate to the geometric interpretation of the simplex method?
 - a. Basic variables define the objective function's slope.
 - b. Non-basic variables represent the constraints' intercepts.
 - c. Each basic feasible solution corresponds to a vertex of the feasible region.
 - d. Basic variables determine the feasible region's boundaries.

- e. Non-basic variables indicate the direction of improvement in the objective function.
5. In the linear program presented on Slide 1, if we set $x_1 = 0$ and $x_2 = 0$, what are the values of the basic variables w_1 , w_2 , and w_3 ?
- a. $w_1 = 12, w_2 = 8, w_3 = 10$
 - b. $w_1 = 0, w_2 = 0, w_3 = 0$
 - c. $w_1 = -12, w_2 = -8, w_3 = -10$
 - d. $w_1 = 1, w_2 = 1, w_3 = 1$
 - e. $w_1 = -3, w_2 = -1, w_3 = 1$
6. On Slide 8, we increase x_1 while keeping $x_2 = 0$. Which constraint determines the maximum feasible value of x_1 ?
- a. $w_1 \geq 0$
 - b. $w_2 \geq 0$
 - c. $w_3 \geq 0$
 - d. $x_1 \geq 0$
 - e. $x_2 \geq 0$
7. On Slide 10, after interchanging x_1 and w_3 , what are the basic variables?
- a. x_1, x_2, w_1
 - b. w_1, w_2, w_3
 - c. x_1, w_1, w_2
 - d. x_1, x_2, w_3
 - e. w_1, w_2, x_2
8. In Slide 11, what is the maximum value of x_2 allowed by the constraints, given that $w_3 = 0$?
- a. 6.8
 - b. 8
 - c. 12
 - d. 2
 - e. 10
9. In the context of the Simplex Method, what is a basic variable, and how does its role relate to the vertices of the feasible region?
- a. A basic variable is a variable that is set to zero in the initial feasible solution, and it corresponds to a point within the feasible region.

- b. A basic variable is a variable that is always non-zero and corresponds to a vertex of the feasible region.
 - c. A basic variable is a variable that is non-zero in a basic feasible solution and corresponds to a vertex of the feasible region; non-basic variables are set to zero at each vertex.
 - d. A basic variable is a variable that is set to one in the initial feasible solution, and it corresponds to a point on the boundary of the feasible region.
 - e. A basic variable is any variable in the problem, and its value determines the location within the feasible region.
10. In the linear program on Slide 1, if we set $x_1 = 0$, what is the value of the basic variable w_2 when $x_2 = 4$?
- a. 4
 - b. 8
 - c. 12
 - d. 0
 - e. 4

2 Answers

1. In the linear program presented on Slide 1, what are the basic variables when x_1 and x_2 are set to 0?
 - a. This answer incorrectly includes x_1 and x_2 as basic variables, even though they are set to 0. The basic variables are those expressed on the left-hand side of the equations.
 - b. When x_1 and x_2 are both 0, the equations on Slide 1 become $w_1 = 12$, $w_2 = 8$, and $w_3 = 10$. Therefore, w_1 , w_2 , and w_3 are the basic variables.
 - c. This answer incorrectly includes x_1 as a basic variable. When $x_1 = 0$, x_1 is a non-basic variable.
 - d. This answer incorrectly includes x_2 as a basic variable. When $x_2 = 0$, x_2 is a non-basic variable.
 - e. This answer incorrectly lists only variables from the original problem statement. Slack variables are also considered in determining basic variables.
2. In the context of the Simplex Method, what does it mean for a variable to be "basic"?
 - a. Non-basic variables are set to zero to obtain a basic feasible solution. Basic variables are not always zero.
 - b. Slack variables are used to convert inequality constraints into equality constraints. Being basic is a property of variables within the simplex method's iterative process.
 - c. A basic variable's value is determined by the values of the non-basic variables. In the simplex tableau, basic variables correspond to columns with an identity matrix structure.
 - d. The coefficient in the objective function influences the selection of entering variables, but it doesn't define whether a variable is basic or not.
 - e. While basic variables are part of the solution at each iteration, they are not guaranteed to be in the final optimal solution. The final optimal solution contains a specific set of basic variables.
3. Why is the concept of basic and non-basic variables crucial in the simplex method?
 - a. While it simplifies calculations at each iteration, the core reason for the distinction is the exploration of vertices.
 - b. By systematically setting non-basic variables to zero and solving for basic variables, the simplex method efficiently moves from one vertex of the feasible region to another, searching for the optimal solution.
 - c. Non-negativity is a constraint of the problem, not a direct consequence of basic/non-basic variables.
 - d. Convergence is a property of the algorithm, but the basic/non-basic distinction is the mechanism that enables it.
 - e. While the simplex method implicitly identifies binding constraints, the core role of basic/non-basic variables is the systematic exploration of vertices.
4. How does the identification of basic and non-basic variables relate to the geometric interpretation of the simplex method?
 - a. The objective function's slope is determined by the coefficients in the objective function, not the basic variables.

- b. Constraints' intercepts are determined by the constants in the constraints, not the non-basic variables.
 - c. Setting non-basic variables to zero and solving for basic variables yields a basic feasible solution, which geometrically corresponds to a vertex of the feasible region. The simplex method iteratively moves between these vertices.
 - d. The feasible region's boundaries are defined by the constraints, not the basic variables.
 - e. The direction of improvement is determined by the coefficients of the non-basic variables in the objective function, but this is not the primary geometric interpretation of basic/non-basic variables.
5. In the linear program presented on Slide 1, if we set $x_1 = 0$ and $x_2 = 0$, what are the values of the basic variables w_1 , w_2 , and w_3 ?
- a. Answer A is correct. Substituting $x_1 = 0$ and $x_2 = 0$ into the equations on Slide 1 gives $w_1 = 12 + 0 - 0 = 12$, $w_2 = 8 - 0 - 0 = 8$, and $w_3 = 10 - 0 + 0 = 10$.
 - b. Answer B is incorrect. This would only be true if the right-hand side of all constraints were 0.
 - c. Answer C is incorrect. The slack variables represent the difference between the left and right sides of the constraints, and they must be non-negative.
 - d. Answer D is incorrect. There is no basis for these values.
 - e. Answer E is incorrect. These values are not consistent with the given constraints.
6. On Slide 8, we increase x_1 while keeping $x_2 = 0$. Which constraint determines the maximum feasible value of x_1 ?
- a. Answer A is incorrect. The constraint $w_1 \geq 0$ leads to $x_1 \geq -12$, which is not a limiting constraint.
 - b. Answer B is incorrect. The constraint $w_2 \geq 0$ leads to $x_1 \leq 8$, which is less restrictive than $x_1 \leq 5$.
 - c. Answer C is correct. The constraints $w_2 \geq 0$ and $w_3 \geq 0$ give $x_1 \leq 8$ and $x_1 \leq 5$ respectively. Since $x_1 \leq 5$ is the more restrictive constraint, $w_3 \geq 0$ determines the maximum feasible value of x_1 .
 - d. Answer D is incorrect. $x_1 \geq 0$ is a non-negativity constraint, but it does not limit the increase of x_1 in this case.
 - e. Answer E is incorrect. Since we are keeping $x_2 = 0$, this constraint is not relevant in determining the maximum value of x_1 .
7. On Slide 10, after interchanging x_1 and w_3 , what are the basic variables?
- a. Answer A is incorrect. x_2 is a non-basic variable at this stage.
 - b. Answer B is incorrect. This is the set of basic variables before the interchange.
 - c. Answer C is correct. After the interchange, x_1 replaces w_3 in the basis, so the basic variables are x_1, w_1, w_2 .
 - d. Answer D is incorrect. w_3 is no longer a basic variable.
 - e. Answer E is incorrect. x_2 is a non-basic variable at this stage.

8. In Slide 11, what is the maximum value of x_2 allowed by the constraints, given that $w_3 = 0$?
- a. Answer A is incorrect. This is the maximum value allowed by the $w_1 \geq 0$ constraint.
 - b. Answer B is incorrect. This value is not derived from any of the constraints.
 - c. Answer C is incorrect. This value is not derived from any of the constraints.
 - d. Answer D is correct. The constraints $w_1 \geq 0$ and $w_2 \geq 0$ give $x_2 \leq 34/5 = 6.8$ and $x_2 \leq 2$ respectively. The smaller value, 2, is the maximum feasible value of x_2 .
 - e. Answer E is incorrect. This value is not derived from any of the constraints.
9. In the context of the Simplex Method, what is a basic variable, and how does its role relate to the vertices of the feasible region?
- a. Incorrect. While some basic variables might be zero in *some* basic feasible solutions, they are not necessarily zero in the initial solution, and a basic feasible solution always corresponds to a vertex, not just a point within the region.
 - b. Incorrect. Basic variables are not always non-zero; they can be zero at certain vertices. Also, the correspondence is to a vertex, not just any point.
 - c. A basic variable is a variable that is non-zero in a basic feasible solution. In the geometric interpretation of the simplex method, each basic feasible solution corresponds to a vertex of the feasible region. The non-basic variables are set to zero at each vertex, defining the coordinates of that vertex in terms of the basic variables.
 - d. Incorrect. Basic variables are not necessarily set to one. The value of a basic variable is determined by the constraints and the values of the non-basic variables.
 - e. Incorrect. Basic variables are a subset of all variables; only the basic variables determine the location of a vertex in the feasible region when the non-basic variables are set to zero.
10. In the linear program on Slide 1, if we set $x_1 = 0$, what is the value of the basic variable w_2 when $x_2 = 4$?
- a. This answer incorrectly assumes w_2 is equal to x_2 .
 - b. This answer incorrectly uses the value of 8 from the equation for w_2 without considering the values of x_1 and x_2 .
 - c. This answer is incorrect; it uses a value unrelated to the problem.
 - d. The equation for w_2 is given by $w_2 = 8 - x_1 - x_2$. Substituting $x_1 = 0$ and $x_2 = 4$, we get $w_2 = 8 - 0 - 4 = 4$.
 - e. This answer is a repeat of option A.